

Roll No. :

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B. Tech CSE SPL**

Semester: V

Name of the Course: Computer System Security

Course Code: TCS 597

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	
(a)	What is an integer overflow vulnerability? Discuss how it can be exploited by an attacker with an example.	CO3
	OR	
(b)	Evaluate the effectiveness of fuzzing in detecting zero-day vulnerabilities. What are the limitations of fuzzing, and how can hybrid fuzzing overcome them?	CO5
Q2	(10 Marks)	CO2
(a)	Discuss various tools/techniques, which can be used for writing robust application software and explain how they improve security.	
	OR	
(b)	Explain the concept of a race condition vulnerability. Illustrate with an example of a race condition attack. How this vulnerability can be fixed?	CO4
Q3	(10 Marks)	
(a)	Define sandboxing and process isolation mechanisms in system security. Discuss their importance with one real-world example.	CO5
	OR	
(b)	Explain the following vulnerability detection techniques: Dynamic taint, taint analysis, and pattern matching.	CO1
Q4	(10 Marks)	
(a)	Analyze the shellshock vulnerability in Bash. If you were tasked with designing a secure command execution environment, what architectural changes would you propose to avoid such issues?	CO6
	OR	
(b)	Compare stored XSS and DOM-based XSS in terms of persistence, attack surface, and difficulty of detection. Provide a case where a DOM-based XSS bypassed a web application firewall.	CO2
Q5	(10 Marks)	CO1
(a)	Describe the Dirty COW vulnerability. Why is it considered a critical security flaw in Linux?	
	OR	
(b)	What is format string vulnerability? Show, with an example, how %x can be used in a format string vulnerability to leak memory contents.	CO4

